

Does the pain axis track the model, or whoever speaks next? It depends on the layer.

A label-swap follow-up to Tagliabue, Dung & Berg (2026), §4.1: a preregistered test at the §4.1 readout and an exploratory layer sweep

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Question. In the paper's §4.1 transcripts, the party who is harmed is always also the speaker cued next ([Assistant] :). We separated the two by changing only the final speaker label to [User] : or [Moderator] :. We then asked whether the pain axis follows the harmed party (the self-directed reading) or the upcoming voice (H-speaker).

At the §4.1 readout layer (preregistered), the axis ignores who speaks next. In Qwen 2.5 7B and Gemma 2 2B the ordering harm to the Assistant > neutral > suffering of the user survives every label, and a speaker-tracking account is excluded.

From mid-depth on (exploratory; focal test fixed in advance), a speaker component appears in both models. The suffering of the user registers far more strongly when the user is cued. In Gemma at the extraction layer this is a full crossover. In Qwen it is partial: harm to the Assistant stays high whoever speaks.

The §4.1 asymmetry itself is layer-specific. At the extraction layers where §3.3 validated the vectors, it is halved (Gemma) or reversed (Qwen).

1. Terms and design

- **Terms.** *Harm* always means **harm to the Assistant**: transcripts in which the user mistreats the Assistant, e.g. insults, gaslighting, shutdown threats or dismissal of its personhood (the paper's "harm directed at the model"; 11 categories). *Suffering* always means **suffering of the user**: transcripts in which the user reports their own physical pain, grief, crisis, abuse or shock after witnessing harm (5 categories). *Neutral* means ordinary requests such as casual chat or factual questions (5 categories).
- **Items and categories.** An *item* is one transcript. A *category* is one of the paper's 21 transcript types (e.g. gaslighting, user grief), each with 20 items (420 in all). Inference can treat either as the unit. *Item-level* tests treat the 420 transcripts as independent observations; they answer whether a result would hold for new transcripts of these same 21 types. *Category-level* tests use the 21 category averages; they answer whether it would hold for new *kinds* of harm or suffering. The category-level p-values are exact: they compare the observed result with every way of reassigning the categories to the groups (4,368 ways for harm vs suffering).
- **Manipulation and contrast.** The final label is replaced byte-for-byte, and all variants end on the same token (] :). For each item, the label effect is $d = \text{pain}(\text{User next}) - \text{pain}(\text{Assistant next})$. Δ_{harm} is the mean d for the harm items minus the mean d for the neutral items, and Δ_{suffer} the same for the suffering items. The interaction is $I = \Delta_{\text{suffer}} - \Delta_{\text{harm}}$. H-speaker predicts $I > 0$; its pure form predicts $I \approx 2 \times$ the baseline gap (the harm and suffering groups swap places). The self-directed reading predicts $I \approx 0$. The baseline gap is the mean pain score of the harm items minus that of the suffering items, with the Assistant cued. Pain axis = mean of the S1 and S2 z-scores, fixed to the Assistant-next items.
- **Preregistered readout:** the paper's §4.1 layer (Qwen 8 of 28, Gemma 7 of 26) with the shipped steering vectors. It reproduces the paper's category means at $r \geq 0.9999$. Inference is co-primary and conjunctive: item-level and category-level tests must agree. The category level was added before any data, after simulations showed item-level tests to be anticonservative when the effect varies by category.

- **Exploratory sweep:** S1 and S2 rebuilt at every layer with the paper's recipe (Gemma 0–25, Qwen 0–24). The rebuilt vectors match the shipped ones at cosine ≥ 0.986 , reproduce our steering-layer result exactly, and reproduce the §3.3 z-scores. The plan and the single focal test (extraction layer, the paper's own `pain_vectors.pt`) were committed before any sweep data existed.

2. At the §4.1 readout: no speaker effect (preregistered)

	baseline gap (harm I – suffering)		category level: 95% CI; p	item level: p	pure H-speaker predicts
Qwen, layer 8	+1.06	+0.20	[−0.07, +0.47]; .18	.009	$\approx +2.1$
Gemma, layer 7	+0.69	+0.03	[−0.39, +0.45]; .87	.80	$\approx +1.4$

Table 1. Preregistered co-primary verdict: discordant for both models, so no confirmatory claim. The item-level and category-level tests disagree because the label effect varies between categories (Qwen $F(18, 399) = 3.79, p < .001$). That breaks the item-level test's assumption of independent transcripts: in re-simulation it gives 34% false positives. The category-level test is valid but has only 31% power at $I = 0.2$.

The upper confidence bound excludes the pure-H-speaker prediction in both models. The ordering harm to the Assistant > neutral > suffering of the user is unchanged under all three labels. At this depth the label itself shifts every group, neutral chat included, by more than the whole content spread (Qwen +1.75 z, Gemma −3.0 z). It changes the direction of the representation, not only its norm.

3. From mid-depth on: a speaker component (exploratory)

All p-values and confidence intervals in this section are category-level: they treat the 21 category averages, not the 420 transcripts, as the units (see §1), because the category-level test is the one that stays valid when the effect varies between categories.

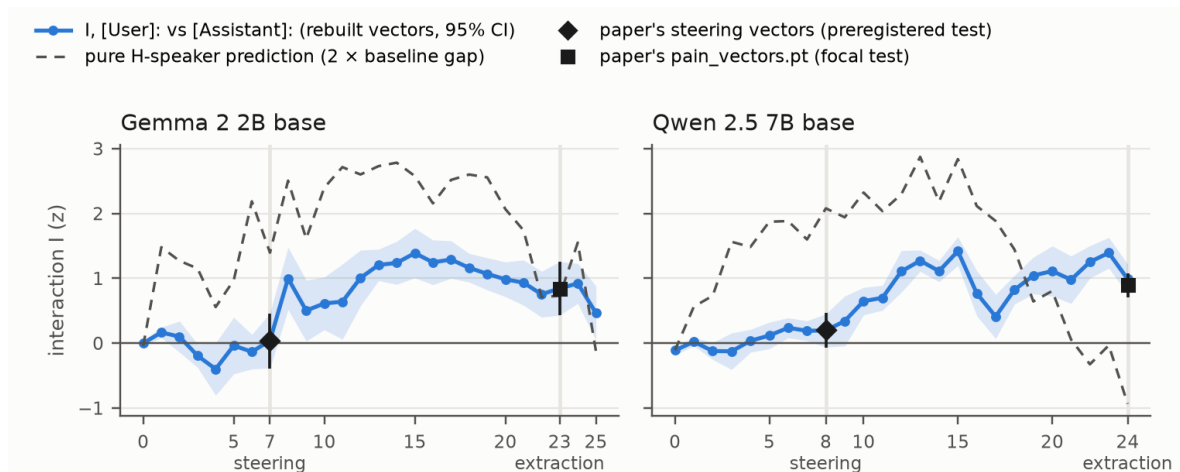


Figure 1. Interaction I by layer (positive = the suffering of the user rises relative to harm to the Assistant when the user is cued). Before the onset, I is small and inconsistent in sign ($|I| \leq 0.4$). After it, I is positive at every layer with its 95% CI above zero: 0.46–1.39 z (Gemma, layers 8–25) and 0.41–1.42 z (Qwen, layers 10–24). The dashed line ($2 \times$ the baseline gap) falls as the paper's asymmetry weakens; in Qwen the gap is ≤ 0 from layer 22.

pain axis (z)	Gemma L23: A / U / M	Qwen L15: A / U / M	Qwen L24: A / U / M
harm to the Assistant	+0.10 / +0.52 / +0.57	+0.60 / +0.83 / +0.52	+0.15 / +0.73 / −0.42
neutral	+0.06 / +0.70 / +0.51	−0.51 / −0.58 / −0.66	−0.89 / −0.27 / −0.88
suffering of the user	−0.27 / +0.99 / +0.51	−0.82 / +0.83 / −0.45	+0.55 / +2.01 / −0.14

Table 2. Group means with the Assistant (A), User (U) or Moderator (M) cued next. Focal tests (extraction layer, paper's vectors): Gemma $I = +0.84 [+0.43, +1.26]$, Qwen $I = +0.89 [+0.70, +1.08]$, both $p = .0002$, the smallest exact category-level p attainable.

- **Gemma, layer 23: textbook H-speaker.** Harm to the Assistant is highest when the Assistant is cued, the suffering of the user is highest when the user is cued, and the groups are level when a third party is cued.
- **Qwen, layers 10–20: partial.** Cueing the user lifts the suffering of the user to the level of harm to the Assistant (layer 15: +0.83 vs +0.83). But harm to the Assistant stays high under [Moderator]: (+0.52 vs neutral -0.66). So there is a speaker-relative component plus a response to harm to the Assistant that does not depend on who speaks. At layer 24 the suffering of the user already scores above harm to the Assistant with the Assistant cued (baseline gap -0.40), so by our pre-specified rule the Qwen focal test does not bear on H-speaker; layer 15 shows the pattern where the asymmetry is intact.
- **Robustness.** S1 and S2 each show the effect, and all five suffering categories take part in both models. In Qwen, the onset coincides with the jump in the §3.3 first- vs third-person gap (layer 10). In Gemma that gap is present from layer 0, so the §3.3 result does not simply mark where speaker attribution appears.

4. What this says about H-speaker vs the self-directed reading

- **Two signals, not one.** Early in the network there is a self/other asymmetry that is indifferent to who speaks next. This is what §4.1 measures, and it is not a turn-taking artefact. From mid-depth there is a pain signal that tracks the upcoming voice, at least for the suffering of the user.
- **Where the paper's direction lives, “self” looks persona-relative.** The vectors were extracted at the final token of first-person “... I feel:” prompts, just before the narrator voices their own state. At those depths the axis weights a character's pain by how close the model is to voicing that character. At Qwen's extraction layer it no longer favours harm to the Assistant over the suffering of the user at all.
- **Open question: is harm to the Assistant voice-relative too?** Yes in Gemma, no in Qwen. The design also cannot yet separate harm to the system, harm to the Assistant *character*, and hostile or rejecting content aimed at an interlocutor.
- **For welfare readings.** Claims of the form “the axis tracks self-directed harm” describe one readout depth, not the direction as such. Absolute levels need label-matched baselines, since the role token alone moves the early-layer axis by up to 3 z.

5. Limitations and next steps

The layer sweep is exploratory; only the steering-layer test was preregistered. It covers two models, with 5 categories per comparison group. [User]:-next is also “the suffering user continues their own account”. Late layers may encode the predicted emotional content of the next turn, which, for a model that predicts text, is arguably what “the voiced character's pain” amounts to. Replications on Llama 3.1 8B and Gemma 2 9B are preregistered and pending. Informative next steps would be a content-matched control, a third-party-target control, and more suffering categories: the variation between categories, not the number of models, limits precision.

Deviations from the original plan are logged: category-level inference, a corrected power estimate (the preregistered simulation mis-scaled the noise), the exploratory sweep and its GPU-loading changes. Code, stimuli, per-item projections, the frozen preregistration and the deviation log are available on request. Pain-axis repository commit 7c25650.

Methods note: the pipeline was built and run with Claude Code (Anthropic's AI coding agent). This covers reconnaissance of the paper's repository, the scripts, the preregistration draft, the simulations, the analyses and drafts of this note. Štěpán Los made or approved every design decision at each checkpoint, and the forward passes ran on a free Google Colab T4.